

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Digital Twin for Well-to-Surface Optimization of Cyclic Steam Stimulation (CSS) and Sucker
Rod Pump (SRP) Operations
Heavy Oil Wells – Baghewala Field, Rajasthan

Smart India Hackathon (SIH) 2026
Problem Statement Implementation Spec

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Contents

1	Executive Summary & Strategic Vision	2
1.1	Context & Problem Context	2
1.2	Product Vision	2
2	Target Personas & Stakeholders	2
3	System Architecture Scope & Boundary	2
4	Core Functional Requirements (FRs)	2
4.1	FR-1: Real-Time Data Ingestion & Telemetry Processing	2
4.2	FR-2: Coupled Reservoir Thermal & Fluid Viscosity Engine	3
4.3	FR-3: Downhole Mechanical Wave Mechanics & Dyno Card Reconstruction	3
4.4	FR-4: AI-Enabled Dynamic Optimization & Dynamic Speed Control	3
4.5	FR-5: Failure Diagnostics & Anomaly Detection	3
5	Non-Functional Requirements (NFRs)	4
6	Mathematical Specifications & Formulations	4
7	Success Metrics & Target Indicators	4
8	Assumptions, Dependencies & Risk Management	4
8.1	Assumptions	4
8.2	Risks & Mitigations	4

1 Executive Summary & Strategic Vision

1.1 Context & Problem Context

The Baghewala Field in Rajasthan produces ultra-heavy crude oil (17°–19° API) from the tight Jodhpur Sandstone reservoir. The reservoir exhibits severe operational challenges:

- High fluid viscosity (> 66 cP at initial 47°C reservoir temperature, surging exponentially as the well cools).
- High asphaltene content causing structural fluid resistance and rapid deposition.
- Low reservoir energy (low natural pressure, 46–48°C native temperature).
- Poor primary oil mobility requiring thermal Enhanced Oil Recovery (EOR) via Cyclic Steam Stimulation (CSS).

Currently, field operations manage CSS cycles (Huff-Soak-Puff) and Sucker Rod Pump (SRP) artificial lift systems in decoupled operational silos based on static historical rules of thumb. As steam condensates and the reservoir temperature degrades post-injection, fluid viscosity surges. This leads to heavy viscous drag (F_{drag}), severe rod stretch (ΔL), rod floating on downstrokes, impact loading, pump unsetting, high failure rates, increased energy consumption, and poor Steam-Oil Ratio (SOR).

1.2 Product Vision

The goal is to develop an integrated, physics-informed, AI-enabled **Well-to-Surface Digital Twin**. This system will continuously couple reservoir thermodynamic decay, dynamic fluid viscosity shifts, downhole sucker rod dynamic wave mechanics, and surface Variable Frequency Drive (VFD) controls. The platform provides real-time predictive monitoring, closed-loop parameter optimization, and failure prevention.

2 Target Personas & Stakeholders

3 System Architecture Scope & Boundary

The Digital Twin operates across three coupled domains:

1. **Reservoir Subsurface Domain:** Continuous modeling of thermal energy delivery, heat dissipation, temperature decay $T(t)$, and empirical dynamic fluid viscosity $\mu(t)$.
 2. **Wellbore & Downhole Mechanical Domain:** 1D damped wave dynamics of the elastic rod string, fluid drag calculation, rod stretch (ΔL), effective stroke loss, dynamic load reconstruction ($PPRL$), and downhole dynamometer card synthesis.
 3. **Surface Automation & Control Domain:** Real-time SCADA/VFD telemetry processing, dynamic SPM_{safe} calculation, automated speed control, and economic production cycle termination triggers.
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4 Core Functional Requirements (FRs)

4.1 FR-1: Real-Time Data Ingestion & Telemetry Processing

- **FR-1.1:** Ingest high-frequency SCADA/VFD telemetry including motor current, RPM, Polish Rod Position $x(t)$, Polish Rod Load $F(t)$, wellhead pressure, and wellhead temperature at ≥ 10 Hz.
- **FR-1.2:** Ingest historical offline data: CSS cycle records (steam mass m_s , quality x_{qual} , injection pressure/temp), reservoir core properties (ϕ, S_o, S_w), and failure logs.
- **FR-1.3:** Perform signal filtering (Butterworth/Savitzky-Golay) on raw load-position data to eliminate high-frequency sensor noise prior to dyno card processing.

4.2 FR-2: Coupled Reservoir Thermal & Fluid Viscosity Engine

- **FR-2.1 Thermal Injection Modeling:** Compute net enthalpy delivered to formation (Q_{inj}) during the Huff stage and calculate initial peak temperature T_{steam} .
- **FR-2.2 Thermal Decay Prediction:** Continuously predict spatial/temporal temperature decay $T(t)$ during the Puff phase using the lumped-capacitance heat loss formulation:

$$T(t) = T_{\text{res}} + (T_{\text{steam}} - T_{\text{res}})e^{-\alpha t} \quad (1)$$

- **FR-2.3 Viscosity Dynamic Derivation:** Compute real-time fluid dynamic viscosity $\mu(t)$ using Andrade/Beggs-Robinson empirical equations based on $T(t)$ and 18° API crude parameters.

4.3 FR-3: Downhole Mechanical Wave Mechanics & Dyno Card Reconstruction

- **FR-3.1 Surface Dyno Processing:** Construct surface Dynamometer Cards (F_{surface} vs. Position) in real-time.
- **FR-3.2 Downhole Card Reconstruction:** Solve the governing 1D Damped Wave Equation (Gibbs Model) using numerical finite difference methods to compute downhole plunger displacement $u(0, t)$ and force $F(0, t)$:

$$\frac{\partial^2 u(x, t)}{\partial t^2} = a^2 \frac{\partial^2 u(x, t)}{\partial x^2} - c(t) \frac{\partial u(x, t)}{\partial t} \quad (2)$$

where $c(t) = f(\mu(t))$ represents fluid viscosity-dependent damping.

- **FR-3.3 Mechanical Metrics Engine:** Calculate Rod Stretch ($\Delta L = \frac{F_{\text{load}} L}{AE}$), Peak Polished Rod Load ($PPRL = W_r + W_f + F_{\text{drag}}$), and net effective plunger stroke length $S_{\text{eff}} = S - \Delta L$.

4.4 FR-4: AI-Enabled Dynamic Optimization & Dynamic Speed Control

- **FR-4.1 Dynamic Safe SPM Calculation:** Continuously evaluate maximum non-floating motor pumping speed $SPM_{\text{safe}}(t)$ based on downward buoyant force vs. viscous drag:

$$SPM_{\text{safe}}(t) \leq K \cdot \frac{(\rho_{\text{steel}} - \rho_{\text{fluid}})g}{\mu(t)} \quad (3)$$

- **FR-4.2 Automated VFD Control:** Automatically generate optimal VFD target speed commands, throttling down SPM dynamically as $T(t)$ drops and $\mu(t)$ rises to prevent rod slack and floating.
- **FR-4.3 CSS Cycle Cutoff Trigger:** Calculate real-time cumulative cycle profit rate dJ/dt . Automatically trigger the termination of the production phase (Puff) and prompt the next steam injection cycle when:

$$\frac{dJ}{dt} = P_{\text{oil}} q_o(t) - C_{\text{elec}} P_{\text{motor}}(t) - C_{\text{wear}} \sigma_{\text{rod}}(t) \leq 0 \quad (4)$$

4.5 FR-5: Failure Diagnostics & Anomaly Detection

- **FR-5.1 Rod Floating Alert:** Detect rod float condition when compressive axial loads ($\sigma_{\text{rod}} < 0$) are observed near the top of downstroke.
- **FR-5.2 Diagnostic Classification:** Use AI dynamic card classification (CNN / XGBoost) to classify: Fluid Pound, Gas Locking, Tubing Leak, Mechanical Unsetting, and Heavy Viscous Drag.

5 Non-Functional Requirements (NFRs)

- **NFR-1 Real-time Latency:** Ingestion to dynamic dyno card reconstruction and VFD command generation must execute in < 1.5 seconds per stroke.
 - **NFR-2 System Scalability:** Architecture must scale horizontally to process up to 200 heavy oil wells concurrently in Baghewala asset fields.
 - **NFR-3 Availability & Reliability:** 99.9% system uptime with edge-computing fail-safe mechanisms; if cloud communication fails, local VFD reverts to last known safe SPM_{safe} baseline.
 - **NFR-4 Data Security:** End-to-end encryption (TLS 1.3) for field-to-cloud telemetry transmission; role-based access control (RBAC) enforcing operational write limits.
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6 Mathematical Specifications & Formulations

7 Success Metrics & Target Indicators

8 Assumptions, Dependencies & Risk Management

8.1 Assumptions

- Wellhead and downhole telemetry sensors remain calibrated within a $\pm 2\%$ error margin.
- Crude API gravity stays consistent at 18° API throughout the primary drainage radius.

8.2 Risks & Mitigations

- **Risk:** High latency or failure in field cellular network during VFD closed-loop adjustment.
 - *Mitigation:* Deploy an Edge Gateway controller running a micro-compiled C++ instance of SPM_{safe} locally at the wellsite.
- **Risk:** Convergence issues during numerical solution of 1D wave equation under extreme dynamic damping.
 - *Mitigation:* Implement implicit finite-difference schemes with automated boundary relaxation.